

Balanced localisation independently confirms a distributed carry signal

Output-digit conditioning separates carry status from the digit being predicted

A Balanced discovery

32 carry + 32 no-carry discovery pairs

centre activations within predicted tens digit

↓
57,344 SAE latents

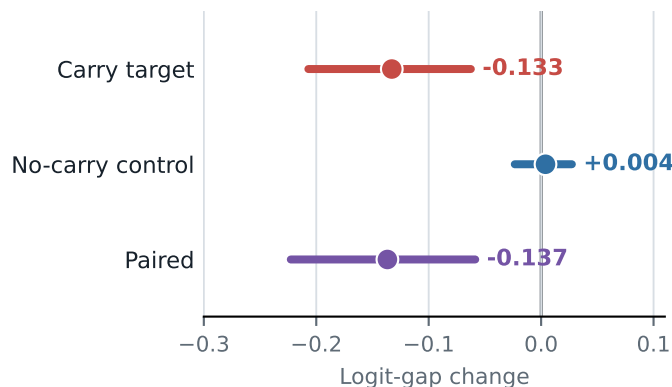
261 pass fixed activity and stratum filters

Cross-protocol convergence

L24F2070 and L28F1204 recur in both graph-ranked and activation-ranked Top-10 panels.

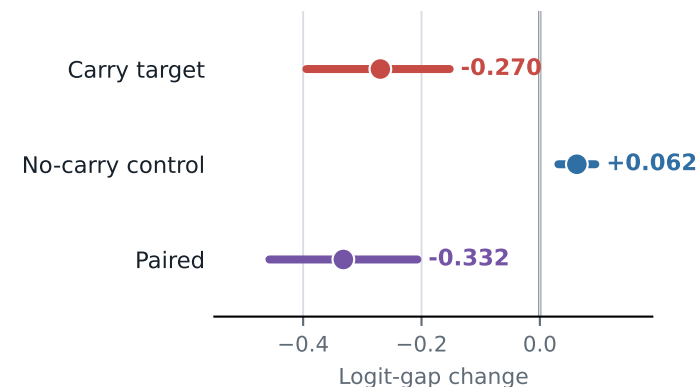
B Frozen Top-10 primary

Confirmation AUC **1.000**
standardised carry difference **+1.095**



C Frozen Top-20 replication

32 intervention-untouched pairs
no reranking; one panel; one test



Top-10 primary: paired effect -0.137, 95% CI [-0.223, -0.059]

Frozen Top-20 replication: paired effect -0.332, 95% CI [-0.457, -0.207]

Interpretation: carry status is strongly represented and selected coordinates make a reproducible causal contribution. No intervention changed the top digit, and one random Top-10 panel was also significant.

The evidence supports a distributed carry-associated panel, not a unique or sufficient carry circuit.